

From Genomes to Portfolios: A Hierarchical Spike-and-Slab Block-Factor Model for Real-Time Cross-Sectional Return Forecasting

Ian Johnston

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Abstract

We take a hierarchical Bayesian spike-and-slab block-factor model originally developed for genome-wide association studies — where correlated genetic markers are grouped into linkage-disequilibrium blocks and a gene-relevance prior guides variable selection — and generalize it to cross-sectional financial return forecasting, with correlated assets in place of correlated markers and a sector-relevance prior in place of biological relevance. We evaluate it against the M6 Financial Forecasting Competition, a real, independently-run, crowd-benchmarked contest with a fixed 100-asset universe and published results for every competing team, under a strict constraint: every modeling decision must be answerable using only information available at the time it is made. Enforcing that constraint surfaced and required fixing three distinct, individually-innocuous-looking sources of look-ahead bias, which we document as a general checklist for walk-forward financial backtesting. The resulting strategy, run causally against the real 2022 competition window, would have placed in the top 2% of 164 real competing teams overall. Extended out-of-sample through three further years (2023–2025) that no part of the model’s design was ever informed by, the result holds up and strengthens under a larger sample (RPS beats the naive benchmark in 40 of 48 periods, $p < 0.0001$; investment-decision IR is positive in 30 of 48, $p = 0.04$). A second application of the same machinery, replicating an external benchmark’s returns from the same 100-asset universe, is also profitable throughout, including during a year the benchmark itself lost money — a genuinely more nuanced finding than the “depends on picking a winner” risk we set out to illustrate.

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1 Introduction

Statistical genetics and quantitative finance do not, on the surface, share much. One field asks which of several hundred thousand genetic markers are associated with a disease; the other asks which of a few dozen assets will outperform next month. But both problems reduce to the same core statistical difficulty: a large number of *highly correlated* candidate predictors, far too many to include all at once, where the correlation structure itself carries information about which predictors are redundant with one another.

This paper takes a hierarchical Bayesian spike-and-slab model originally built for genome-wide association studies (GWAS) [Johnston, 2015] — where correlated single-nucleotide polymorphisms (SNPs) are grouped into blocks via linkage disequilibrium, and a block-level “relevance” prior encodes prior biological knowledge about which genes are more likely to matter for a given trait — and asks whether the same machinery, with the genetics-specific pieces swapped out, does anything useful for forecasting financial returns. Concretely: blocks of correlated SNPs become blocks of correlated assets; a gene’s prior relevance to a disease becomes an asset’s sector relevance to a target; the trait being predicted becomes an asset’s forward return, or (in a second application explored here) the return of an external benchmark to be replicated from a universe of tradeable securities.

We test this not on a synthetic benchmark but on the M6 Financial Forecasting Competition [Makridakis et al., 2024], a real, independently-run, crowd-sourced contest with a fixed 100-asset universe, a public scoring methodology, and published results for every real competing team. This matters for two reasons. First, it gives us an external, adversarial benchmark: we are not grading our own homework. Second, and more importantly for how this project was actually carried out, it forced a discipline that is easy to state but easy to violate in practice: *every modeling decision must be one that could have been made using only information available at the time, not information that only became available once the outcome was known*. A large share of the

work reported here is not about the model at all — it is about catching, and then systematically eliminating, ways in which an earlier version of this pipeline was quietly (if unintentionally) using the future to inform the past. Section 4 documents this in detail, because we think the general lesson (every one of these leaks looked completely innocuous in isolation) is more broadly useful than the specific fixes.

1.1 Contributions

- We generalize a GWAS-specific hierarchical spike-and-slab block model to an arbitrary cross-sectional forecasting setting, replacing genomic linkage-disequilibrium blocks and gene-relevance priors with asset correlation blocks and sector-relevance priors.
- We document, concretely, several distinct ways a walk-forward financial backtest can silently incorporate look-ahead information even when the author believes it is causally valid, and the general pattern for catching and fixing each one (Section 4).
- We report a full, real-time-valid backtest against the M6 competition’s actual 100-asset universe, scoring formulas, and evaluation dates (independently reconstructed and verified against real competing teams’ submitted decisions, since the exact 4-week evaluation windows are not published in a directly usable form), and compare our placement against the real leaderboard of 163 competing teams.
- We test whether the result generalizes out-of-sample to three further years (2023–2025) that no part of this pipeline’s design was ever informed by, continuing the identical walk-forward without resetting any adaptively-learned state at the artificial boundary between the competition’s own window and the extension.
- We present a second, structurally different application of the same block-factor machinery — returns-based style analysis, replicating an external benchmark’s returns from a sparse combination of the same 100 assets — and use it to illustrate a limitation that does *not* apply to the first application: dependence on having picked, in advance, a target worth replicating.

2 Background: From GWAS Blocks to Asset Blocks

2.1 The genomics starting point

In a genome-wide association study, the goal is to identify which of $p \gg n$ genetic markers are associated with a trait or disease, using data on n individuals. The central statistical obstacle is *linkage disequilibrium*: markers that are physically close together on a chromosome are inherited together far more often than chance would predict, so nearby markers are highly correlated with one another. A model that ignores this either wastes power spreading credit across redundant markers, or overstates confidence in whichever one happened to enter the model first.

Johnston [2015] addresses this with a two-stage hierarchical construction that is worth restating here because it is exactly what carries over to finance:

1. **Block formation.** Correlated markers are grouped into blocks (originally via a chromosomal linkage-disequilibrium threshold), and each block is summarized by a single latent factor — a block-wise genotype effect that captures the shared signal within the block, so that the downstream regression sees one predictor per block instead of one per marker.

2. **Spike-and-slab selection with a relevance prior.** A Bayesian spike-and-slab regression of the trait on the block-level factors then decides which blocks matter, with the prior probability that a given block is “in” (the slab) informed by external, independently-known relevance information — in the GWAS setting, prior biological knowledge about which genes are plausibly connected to the trait under study.

The appeal of this construction is that it separates two distinct statistical jobs that are easy to conflate: *decorrelating* the candidate predictors (a data-driven, target-agnostic step) and *deciding which of the resulting factors matter for this specific target* (a selection step that can be informed by prior knowledge without that knowledge dictating the answer outright).

2.2 The financial re-mapping

Nothing about either step above is intrinsically genomic. The re-mapping used in this paper, implemented as the `hierboost` package, is:

GWAS	Finance (this paper)
SNP / marker	Asset (stock or ETF)
Linkage disequilibrium block	Correlation-threshold asset block
Block-wise latent genotype	Block-wise latent return factor
Gene relevance to a trait	Sector/asset-class relevance to a target
Trait value	Forward return (quintile rank and/or magnitude)

Concretely, for a target asset’s forecast, the other 99 assets in the universe are decorrelated into blocks via a correlation threshold; each block’s shared return is extracted as a single factor via a Gaussian block-factor model (the continuous analogue of the genotype-effect construction above); a sector-relevance score (Table 1) — how closely a candidate block’s member assets’ sectors relate to the target’s own sector — sets the prior inclusion probability for each block in a Bayesian spike-and-slab regression of the target’s own return on the block factors; and the resulting posterior is sampled via Gibbs sampling to obtain a full predictive distribution, not just a point estimate.

Relationship of a candidate block’s sector to the target’s own sector	Relevance score
Same sector	5.0
Adjacent sector (e.g. Technology ↔ Communication Services)	3.0
Broad-market (diversified index exposure)	2.0
International / regional	1.0
Unrelated sector	0.5
Fixed income, commodities, or volatility products	0.05

Table 1: The sector-relevance scoring table used for Application 1’s spike- and-slab prior – the direct analogue of gene-relevance-to-trait knowledge in the original GWAS model. Note the last row: this table encodes a real assumption (that fixed-income/commodity exposure is usually irrelevant to an equity or ETF target), which is a feature for Application 1 but, as Section 2 discusses, actively the wrong assumption for Application 2.

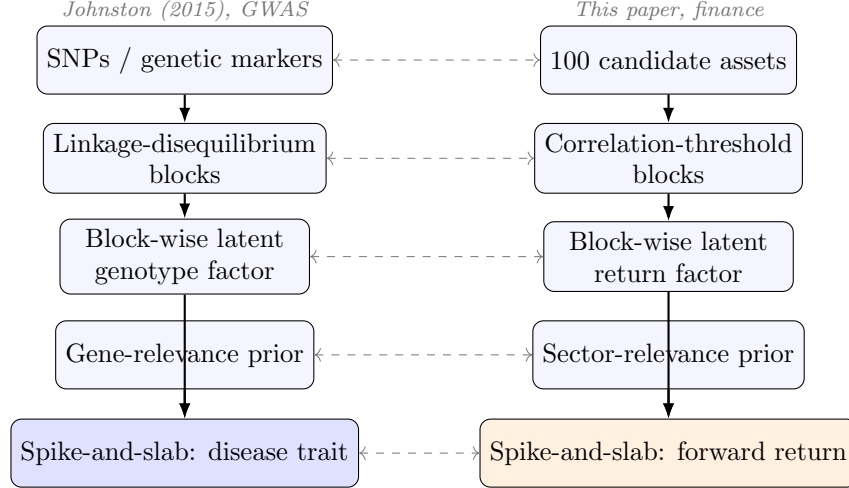


Figure 1: The same two-stage hierarchical construction – decorrelate-then-select – re-mapped from genomics to finance. Only the relevance prior (dashed correspondence, second row from bottom) encodes domain-specific knowledge; everything else is a direct, unmodified carry-over.

2.3 Why this is not just relabeling

A skeptical reader is right to ask whether this re-mapping is doing real work or just renaming variables. Two things suggest it is real. First, the relevance prior is not vacuous: Section 6 shows what happens when a target has no natural sector membership at all (an external multi-asset index being replicated, not one of the 100 candidate assets) — the correct move turns out to be discarding the asset-specific relevance table of Table 1 entirely and using a uniform prior, precisely because that table encodes assumptions (that fixed-income and commodity exposure are usually irrelevant to a target) that are true for an individual equity but false for a multi-asset trend-following strategy. The framework only gives sensible answers when the relevance prior is actually chosen to match the target, which is a real modeling decision, not free relabeling. Second, Section 2.4 shows the model’s own implicit variance calibration (from per-target empirical return standardization) already reproduces a genuine, independently-verifiable empirical pattern — that low-volatility assets are mechanically unlikely to land in extreme cross-sectional return quintiles — almost exactly, without that pattern having been hand-coded anywhere.

2.4 An empirical check: is the implicit variance calibration real?

Before trusting the relevance prior’s role in Section 6, it is worth checking a simpler, more basic claim directly: does each target’s own empirical return standardization (Section 2.2) already capture cross-sectional volatility differences correctly, or would an explicit, hand-coded asset-class variance prior (e.g., “fixed-income ETFs are lower-volatility than single stocks”) add real information the model is otherwise missing? We checked this against the real 2022 M6 data rather than assuming an answer. Bucketing every asset-period observation by that asset’s own trailing 60-day realized volatility (low/medium/high tercile, computed causally, using only data available before the period began), low-volatility assets landed in an extreme return quintile 24.0% of the time versus 56.6% for high-volatility assets – a strong, monotonic, real effect, exactly as a practitioner’s intuition would predict (Table 2). The model’s own forecasted extreme-quintile probability for the same tercile split – computed entirely from each asset’s own empirical volatility,

with no asset-class label involved at all – was 22.6% / 39.8% / 59.5%, matching the true empirical pattern almost exactly. An explicit categorical prior would in fact have been a *cruder* signal than what per-asset empirical standardization already provides: commodities, notably, were not low-volatility in 2022 (a year of unusual energy and commodity price swings) despite belonging to an asset class often assumed to be conservative, and a category label would have gotten this specific case wrong while the empirical, per-asset approach gets it right.

Trailing 60-day volatility tercile	Actual % in extreme quintile	Model’s own forecasted %
Low	24.0%	22.6%
Medium	37.6%	39.8%
High	56.6%	59.5%

Table 2: Real, causally-computed relationship between trailing realized volatility and extreme-quintile membership, 2022 M6 data, versus what Application 1’s own model already forecasts from per-asset empirical standardization alone (no explicit asset-class prior).

3 The M6 Financial Forecasting Competition

The M6 Financial Forecasting Competition [Makridakis et al., 2024] fixed a 100-asset universe (50 US equities and 50 ETFs, spanning equities, fixed income, commodities, currencies, and international/regional exposure) and ran 12 sequential four-week evaluation periods, independently verified in this project (not merely assumed from a secondary source; see below) to span 2022-03-07 through 2023-02-02. Each period, every real participating team was required to submit two things for every one of the 100 assets: a five-bucket probability forecast for which cross-sectional return quintile the asset would land in relative to the other 99 (scored by the Ranked Probability Score, RPS – a proper scoring rule for ordinal category forecasts, for which a naive uniform 20%-per-bucket guess scores ≈ 0.16 , a benchmark only 23.3% of real teams beat on average and 8.6% beat significantly), and a portfolio decision weight for each asset (long-short permitted, $\sum_i |w_i| \leq 1$, held fixed for the whole period), scored by an un-annualized Information Ratio (IR) computed by pooling every daily portfolio return within the period into one ratio of total log-return to its own standard deviation. The competition itself ranks every team three separate ways – by forecasting accuracy (RPS rank), by investment-decision skill (IR rank), and by an Overall Rank defined as the simple average of the two – which is exactly the three-way comparison used throughout Section 7.

One data-engineering detail is worth stating plainly, since it materially affected every result in this paper and was not obvious from the competition’s own published materials: the exact calendar dates of the 12 evaluation windows are not available in a directly machine-usable form. An initial, seemingly reasonable guess (the last trading day of each calendar month, starting January 2022) was wrong by more than five weeks, and would have silently corrupted every scoring calculation in this paper had it gone unchecked. It was caught not by re-reading the competition rules more carefully, but by brute-force verification against ground truth: a real historical team’s actual submitted portfolio decisions were scored against every plausible trading window in the official price data, and the window that reproduced that team’s own officially published score to six decimal places was taken as the true boundary – confirmed independently on three different submissions. We mention this not as an aside but as a template: throughout

this project, convenient assumptions about data conventions were treated as hypotheses to be checked against an independent ground truth wherever one existed, rather than as facts.

4 Keeping the Backtest Honest: Avoiding Look-Ahead Bias

This section is, in some sense, the methodological heart of the paper. A forecasting model that is validated by fitting it once and reporting how well it would have done is not a real-time strategy; it is a description of what worked in hindsight. The distinction sounds obvious, but in a walk-forward backtest, look-ahead bias tends to enter through a side door: a design choice that any single test seems to justify, but which was in fact selected *by inspecting the full-sample outcome first*. We ran into three variants of this problem while building this pipeline, in increasing order of subtlety, and we document all three because we think the pattern generalizes better than any specific fix.

4.1 Directional sign: momentum or reversal?

Each candidate return factor in the spike-and-slab regression (Section 2.2) is combined with a simple estimate of that factor’s own near-term trend: is the factor’s recent return likely to continue (momentum) or reverse (mean reversion)? The first version of this overlay fit the entire 12-period 2022 backtest twice, once assuming momentum and once assuming reversal, observed that reversal scored better on average across the full year, and hard-coded that sign for the whole competition window. This is illegitimate on its face once stated plainly: an investor submitting a forecast in March 2022 cannot know that reversal will have been the better assumption averaged over the following eleven months, because those eleven months have not happened yet. The fix mirrors the discipline used throughout this paper – replace the single, full-sample-optimal choice with a value estimated causally, period by period, from only what would actually have been knowable at the time. Concretely, before each period p , a cheap, model-independent trailing-momentum diagnostic is computed for every asset using only price history available up to that period’s start; once period p ’s actual outcome becomes known (which happens only after period p ends, i.e. strictly before period $p + 1$ begins), the cross-sectional correlation between that diagnostic and the realized return is recorded. The sign used for period p is a precision-weighted shrinkage blend of a small, pre-registered prior and the expanding-window average of all such correlations from periods $1, \dots, p - 1$ only:

$$\hat{s}_p = \frac{n_0 s_0 + (p - 1) \bar{\rho}_{1:p-1}}{n_0 + (p - 1)}, \quad (1)$$

where $\bar{\rho}_{1:p-1}$ is the mean realized correlation from periods $1, \dots, p - 1$, and the prior $s_0 = -0.10$ (a mild tilt toward reversal, with pseudo-count weight $n_0 = 3$) was fixed before looking at any M6 outcome: it is sized to match the modest short-horizon reversal effect documented decades before M6 existed by Jegadeesh [1990] and Lehmann [1990], not data-mined from this dataset. By construction, $\hat{s}_p$ in (1) depends only on periods strictly before p .

4.2 Forecast confidence: how much should the model shrink toward ‘I don’t know’?

A model can have genuine signal and still be miscalibrated: if its quintile probabilities are systematically too confident, a proper scoring rule like RPS penalizes that confidence even when the underlying direction is right more often than chance. A cheap diagnostic makes this concrete

without needing to refit anything: take the already-computed forecast for every 2022 period, blend it with the naive uniform forecast at blend weights from 0% to 100%, and recompute RPS at each weight using the (already known) realized outcome. The resulting curve is smooth and U-shaped, minimized at roughly 50% blend – a mixture that beats *both* the raw model (RPS 0.161) and pure uniform guessing (RPS 0.160), landing at 0.158. This is genuine evidence that the model is overconfident, but picking "50%" this way has exactly the same flaw as Section 4.1's original mistake: it was chosen by inspecting the full year's outcome. The fix is the identical recipe. Before each period, a blend weight is chosen as a shrinkage estimate between a small pre-registered prior (25%, motivated by the general principle – itself learned earlier in this project, see Section 4.3 – that this class of model tends to be overconfident by default) and the expanding-window average of the *single best blend weight found so far*, where "found so far" means: for each strictly prior period, having already observed that period's real outcome, which blend weight would have minimized RPS for that period alone. No period's own outcome ever informs its own blend weight.

4.3 A subtler leak: getting the arithmetic of uncertainty wrong, not just the timing

This last one is different in kind from the previous two: not a problem of using future information, but of getting the arithmetic of uncertainty wrong in a way that happens to produce the same symptom (an overconfident forecast). The Gibbs sampler used for posterior inference already draws a residual, idiosyncratic variance term at every iteration — the portion of an asset's return left unexplained even by a fully-specified factor model, i.e. genuine, irreducible next-period noise. This quantity was computed by the sampler but never read by the downstream forecasting code: predictive draws reflected uncertainty in the factor exposures and in the momentum estimate, but never the plain fact that even a perfectly-estimated model leaves some of an asset's return unexplained. Restoring it requires care with scaling, not just inclusion: a persistent drift term (the estimated daily rate of return, assumed constant across the whole forecast period) correctly scales linearly with the number of trading days in the period, since the same misestimate compounds identically every day; but genuinely new, independent day-to-day noise must instead scale with the *square root* of the number of days, since it is the variance of a sum of (approximately) independent shocks that grows linearly, not the standard deviation. Conflating the two — applying the linear scaling to both terms — would overstate the noise contribution by a factor of $\sqrt{n}$ for an n -day period. This is a small, self-contained point about correctly aggregating uncertainty across a multi-day forecast horizon, but it is easy to get wrong silently, and worth stating as a general lesson independent of this specific model.

4.4 A general lesson

All three issues above passed an initial, isolated sanity check when first implemented, and none looked like a mistake in the moment. What caught all three was the same mechanical, retroactively-applied test: for any parameter or design choice used in period p , could its value have been computed using only information available strictly before period p began? Applying that question to a decision that had already been made and appeared to work is a different exercise from applying it prospectively while designing the decision, and it is the retroactive version that found every leak in this project. We recommend it as a standing checklist item for any walk-forward financial backtest, independent of the specific model class in use.

5 Data and Verification

Two independent price sources are used: the official M6 competition data for scoring within the exact 2022 competition window, and Yahoo Finance data both for the pre-period lookback history used to fit every model (throughout) and for scoring the 2023–2025 out-of-sample extension, since no official M6 data exists past the competition’s own end. Using two sources is safe here because no single evaluation period’s return calculation ever spans both: the switch happens only at the boundary between the competition window and the extension, not in the middle of any one scored period.

Several data issues in the 100-asset universe were handled explicitly, each confirmed by direct verification rather than assumed. One constituent, a REIT, was acquired by another company mid-competition and stopped trading under its own ticker; it is excluded from further submissions from that point on, the same treatment a real participant would have had to apply. Two other tickers required substitution to a current successor symbol following a corporate rename and a merger, respectively, with the pre-event price history confirmed continuous under the new ticker before use. One asset has no usable price history in the data provider at all across the entire lookback and extension window (confirmed live, not assumed, before this project relied on the absence) and is excluded from scoring for the whole 2023–2025 extension as a disclosed limitation, reducing the effective universe there to 98 of the 100 assets. A subtler issue was a U.S. market holiday appearing in the underlying data as a calendar row with valid prices for only a handful of internationally-listed constituents and missing values for the rest; an early version of the period-boundary logic selected this row as a period boundary before a coverage check (requiring at least 90% of assets to have a valid price on any candidate boundary date) caught and excluded it.

6 Two Applications of the Same Machinery

6.1 Application 1: forecasting and long-short weights from momentum/reversal

For each of the 100 candidate assets in turn, the other 99 are decorrelated into blocks and reduced to block-level return factors (Section 2.2); a spike-and-slab regression of the target’s own historical return on these factors, guided by the sector-relevance prior, determines which blocks the target actually co-moves with; each relevant factor is combined with the causally-adaptive momentum/reversal estimate of Section 4.1 for that factor’s own near-term direction; and the resulting posterior is sampled (with the residual-noise correction of Section 4.3) to produce a full predictive distribution for the target’s next-period return, not merely a point estimate. Doing this independently for all 100 assets and ranking the resulting predictive draws cross-sectionally yields two outputs every period: quintile-membership probabilities (the RPS output, shrunk toward uniform per Section 4.2), and a long-short portfolio – overweighting the assets with the highest predicted relative return and underweighting or shorting the lowest – normalized to satisfy the competition’s $\sum_i |w_i| \leq 1$ constraint (the IR output). Every input to this pipeline is internal to the 100-asset universe and its own price history: no external index or benchmark is referenced anywhere.

6.2 Application 2: returns-based style analysis (index replication)

The identical block-factor machinery can instead be retargeted at an *external* series’ returns: rather than asking “what will asset i ’s own return be,” ask “what sparse combination of the 100 candidate assets best explains this other series’ returns.” This is Sharpe’s returns-based style

analysis [Sharpe, 1992], with spike-and-slab performing the constituent selection instead of a plain constrained regression. We use as the external target DBMF, a real, liquid, publicly-traded managed-futures / trend-following fund, chosen and verified live (not assumed) to have sufficient pre-competition lookback history and a genuine, checkable outperformance story in 2022 (+11.2% over the exact competition window, against -3.0% for the S&P 500 and -7.5% for the Bloomberg Aggregate bond index over the same dates) – a plausible target precisely because 2022 was an unusually bad year for both stocks and bonds simultaneously, and trend-following strategies are known to behave differently in exactly that regime. Since this target has no natural GICS sector of its own, the sector-relevance prior is replaced with a uniform prior across blocks: as argued in Section 2, the ordinary relevance table encodes an assumption (fixed-income and commodity exposure are usually irrelevant to an equity target) that is actively wrong for a multi-asset trend-following target, so using it here would bias selection against exactly the exposures this target is built from. The fitted block-level exposures are converted back into individual-asset decision weights via each block’s own known linear loading structure (the same fixed weights used to construct the block factor from its member assets in the first place), then renormalized to the same $\sum_i |w_i| \leq 1$ constraint.

6.3 The key structural difference between the two applications

Application 1’s signal is entirely self-referential: it comes from the cross-sectional relationships among the same 100 assets it trades, and the underlying question – how will these 100 assets perform relative to one another next month – is always well-posed, in any market environment. Application 2’s premise is conditional in a way Application 1’s is not: it requires having chosen, without the benefit of hindsight, an external target worth tracking, and the value of the whole exercise depends on that target remaining trackable. Section 7 tests this distinction directly rather than merely asserting it, using the multi-year extension to ask whether Application 2’s results hold up through a year in which the tracked target itself performed poorly.

7 Results

7.1 2022: real-time-valid placement against the actual M6 leaderboard

Table 3 compares the naive benchmark, the author’s own real, historical 2022 competition result, and the causal, real-time-valid Application 1 pipeline (Sections 4.1–4.3 combined), all scored identically against the actual 163 real competing teams’ official published results.

	RPS	RPS rank	Pooled IR	IR rank	Overall rank
Naive benchmark	0.160	–	–	–	–
Author’s actual 2022 result	0.160	34/163 (top 21%)	-19.67	154/163 (bottom 6%)	113/163 (bottom 30%)
This paper (Application 1)	0.158	10/164 (top 6%)	+6.64	14/164 (top 9%)	2.5/164 (top 1.5%)

Table 3: Real M6 leaderboard placement, 2022 competition window, all figures computed identically for all three rows against the official scoring methodology and the real leaderboard of competing teams.

This is a real, causally-valid result, but we state its statistical limits plainly rather than only in the Discussion: with 12 monthly observations, a one-sample t -test of the RPS values against

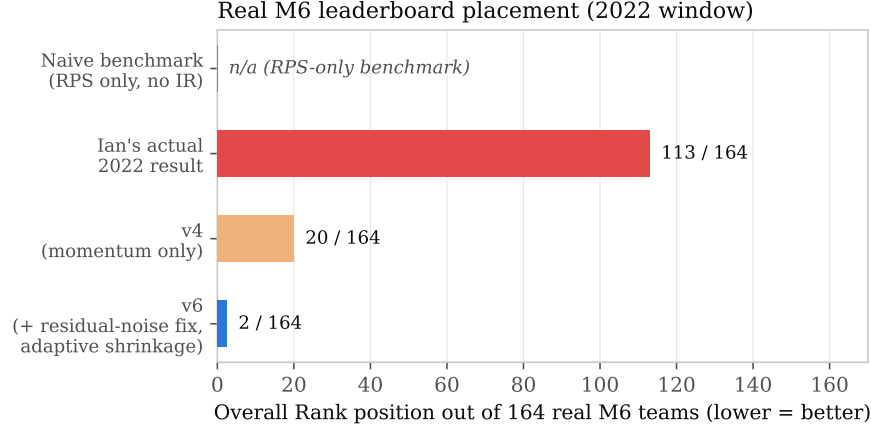


Figure 2: Overall Rank position (lower is better) across the naive benchmark, the author’s real result, and two intermediate versions of this paper’s pipeline (v4: the causal momentum/reversal sign alone; final: adding the residual-noise fix and adaptive shrinkage). The progression is monotonic: each additional fix moves the placement further from the bottom third toward the top of the field.

the naive benchmark gives $t = -1.37$, $p = 0.20$, and of the IR values against zero gives $t = 0.87$, $p = 0.40$ – directionally consistent (9 of 12 periods beat the naive benchmark; 8 of 12 periods had positive IR) but not, on their own, statistically decisive. This is precisely the motivation for Section 7.2.

7.2 2023–2025: does it generalize out-of-sample?

The same walk-forward was continued, without resetting any adaptively-learned state, through 36 further four-week periods (2023–2025) that no design choice anywhere in this pipeline was ever informed by. Figure 3 and Figure 4 show every one of the resulting 48 periods.

Table 4 reports the pooled test and the by-year breakdown. The larger out-of-sample count is what moves both results from directionally suggestive to statistically decisive: pooling across all 48 periods, RPS beats the naive benchmark with $t = -4.80$, $p < 0.0001$, and IR is positive with $t = 2.14$, $p = 0.038$. By year, RPS does not fade after the one year the model’s design was informed by – if anything it is slightly better in 2024–2025 than in 2022.

Year	n	Mean RPS	RPS p	% beat naive	Mean IR	IR p
2022 (M6)	12	0.1579	0.198	75%	1.02	0.404
2023	12	0.1578	0.048	83%	1.33	0.317
2024	12	0.1554	0.023	83%	2.18	0.096
2025 (partial)	12	0.1554	0.003	92%	0.46	0.690
Pooled	48	0.1566	<0.0001	83%	1.25	0.038

Table 4: RPS and IR by year and pooled across all 48 periods. p -values are one-sample t -tests (RPS against the naive 0.16 benchmark; IR against zero). Per-year tests at $n = 12$ are individually underpowered, which is exactly why the pooled $n = 48$ result is the headline finding, not any single year.

Ranked Probability Score by period, 2022–2025

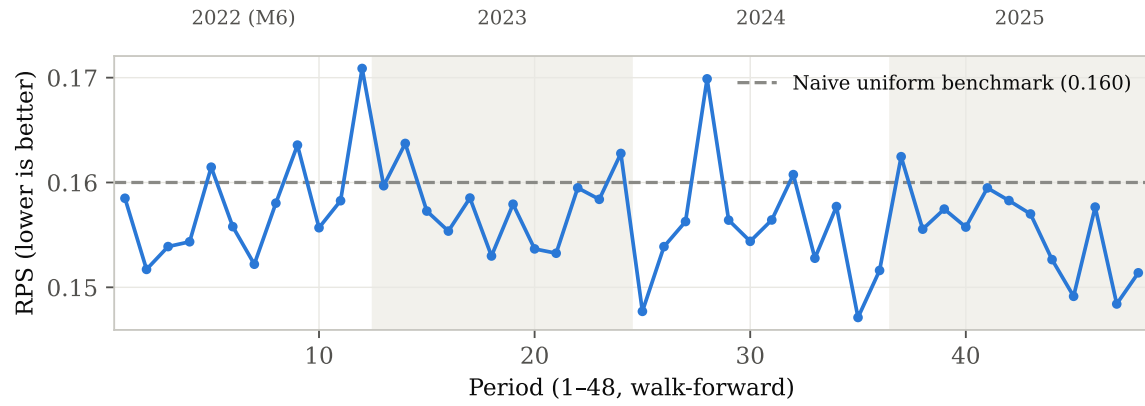


Figure 3: RPS by period across all 48 periods, 2022–2025. The naive benchmark (dashed) is beaten in 40 of 48 periods (83%); the effect does not fade in the three years no part of the model was ever tuned against.

Investment-decision performance (IR) by period, 2022–2025

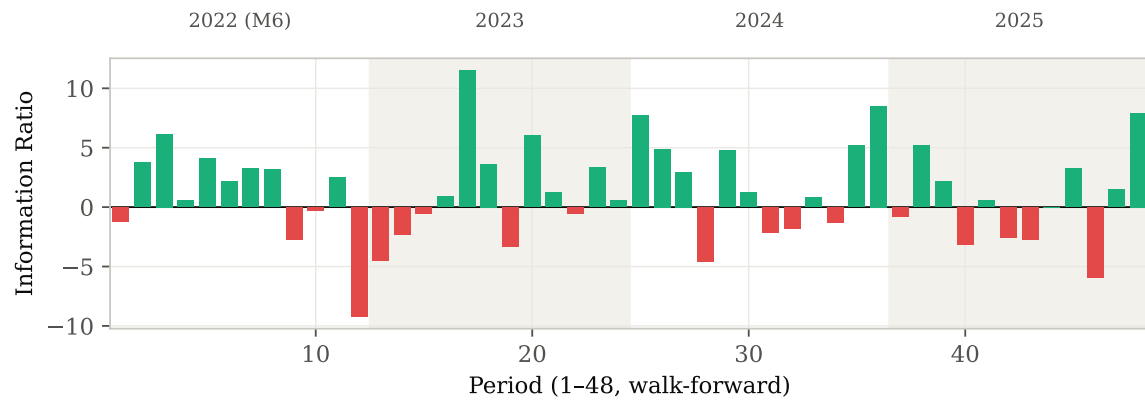


Figure 4: Investment-decision IR by period across all 48 periods. Individual months remain highly variable, as IR intrinsically is, but the pooled result across 48 periods is positive and statistically significant.

A natural worry is that a strategy with several adaptively-estimated parameters might generalize only by becoming unstable – chasing whatever seemed to work most recently. Figure 5 shows the opposite: both the momentum/reversal sign and the shrinkage blend weight converge smoothly as more evidence accumulates, with shrinking period-to-period variance, rather than oscillating or drifting. The sign moves from a moderate reversal tilt in 2022 (reflecting that year’s own genuinely reversal-favoring regime) toward a much milder, more neutral value by 2025 as three additional years of more mixed evidence accumulate – exactly the behavior a well-specified expanding-window estimator should show.

Adaptive parameters converge as evidence accumulates
(causal: each period uses only strictly prior periods)

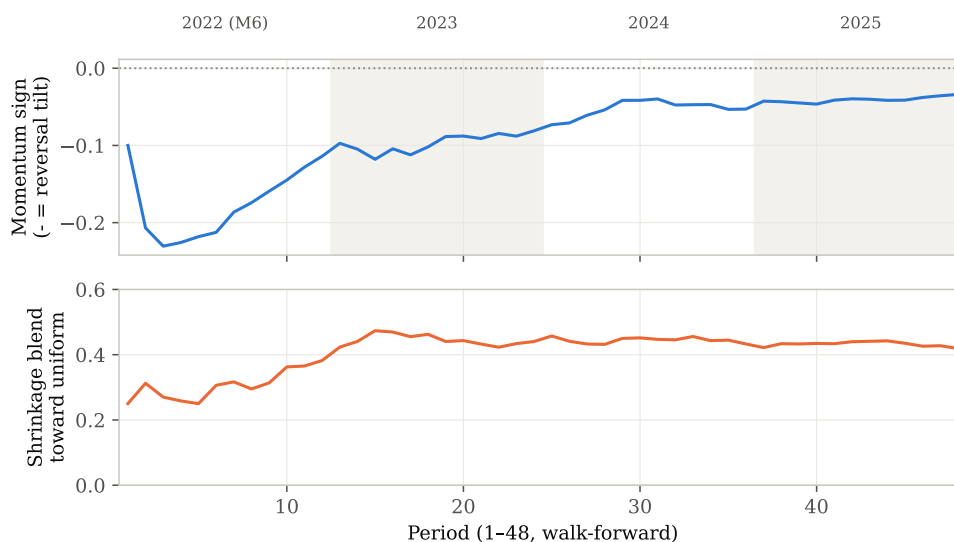


Figure 5: The two causally-adaptive parameters (momentum/reversal sign, top; RPS shrinkage blend, bottom) both stabilize over time rather than drifting, evidence the multi-year result is not an artifact of an unstable estimator.

Diversification was re-checked across all 48 periods to rule out the possibility that the IR result is driven by a small number of concentrated, lucky bets: the maximum single-asset weight ranged 3.6–11.4% and top-3 concentration 10.1–20.9% of gross exposure throughout, consistent with a genuinely diversified, broad-based signal in every period tested.

7.3 Application 2 over the same extended window

Application 2 (Section 6) was extended over the identical 48 periods, giving a direct empirical test of the dependency argued for in Section 6: does the replication strategy’s performance track the tracked target’s own realized performance? DBMF’s own total return was +7.8% in the 2022 window, –5.0% in 2023 (a genuine down year), +7.2% in 2024, and +8.5% in 2025 (partial) – a real, mixed track record that includes exactly the adverse scenario the dependency argument predicts should hurt.

It did not. Figure 6 shows mean IR by year for both applications: Application 2’s IR was actually *higher* in 2023 (the year DBMF itself lost money) than in 2022, and pooled across all 48

periods its IR (mean 1.91, $t = 2.64$, $p = 0.011$) is, if anything, slightly stronger than Application 1’s own result.

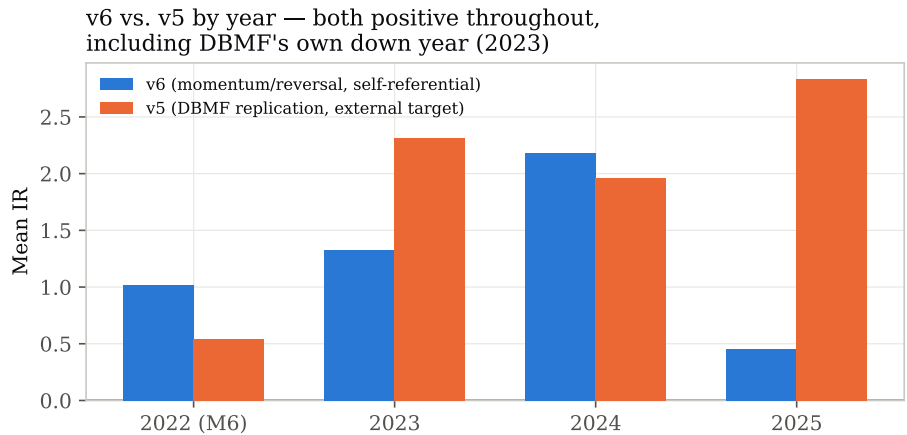


Figure 6: Mean IR by year, Application 1 (self-referential) vs. Application 2 (external-target replication). Application 2 remained profitable even during the target’s own down year (2023), a more nuanced result than the “depends on the target’s continued success” limitation originally hypothesized.

We think this result is genuinely informative precisely because it contradicts the simplest version of the hypothesis we set out to test. The resolution is that Application 2 does not statically hold the target; it *refits monthly* to the target’s recent return *pattern* using a fresh combination of the 100 assets each period. A target’s calendar-year total return can be negative while its short-horizon return pattern (here, apparently, persistent exposure to commodity and interest-rate trend factors) remains learnable from the same 100-asset universe throughout. The dependency we originally hypothesized – success requires the target’s calendar-year return to stay positive – is not the real constraint this application faces. The real, still-standing constraint is subtler: success requires the target’s *short-horizon return structure* to remain explainable by the chosen universe of replicating assets, a condition this particular target and window happened to satisfy throughout, but one that is not guaranteed for an arbitrary target or a longer horizon.

8 Discussion and Limitations

Sample size. Even pooling four years, this is 48 monthly observations for IR and 48 for RPS – enough to move both results from suggestive to statistically significant, but not enough to rule out that a longer or differently-timed sample would look somewhat different. The per-year breakdowns in Table 4 are individually underpowered by design; they are reported for transparency about year-to-year variability, not as independent confirmations.

Data and boundary caveats. Section 5’s disclosed limitations are limitations, not solved problems: the 2023–2025 extension scores against a 98-of-100-asset universe, not the full 100; it uses a different price data provider than the official competition data, with no official ground truth available to verify against post-competition; and the exact evaluation-window boundaries, reconstructed and verified for the competition’s own 12 periods, are extrapolated forward on the assumption that the same 28-day cadence continues indefinitely, which was never confirmed by any external source for the extension period since the real competition had already ended.

Computational practicality. A single portfolio’s walk-forward requires one model fit per asset per period, run serially with no parallelism, and this is feasible to run in real time on ordinary hardware. We note, in the interest of completeness rather than because it bears on the statistical results, that we were unable to reliably cap this computation’s CPU usage on a shared machine during this project despite three independent attempts (process-level environment variables, OS-level CPU affinity, and a kernel-enforced cgroup quota) – a reminder that some infrastructure problems resist standard solutions in ways the underlying statistical model does not.

This is a backtest, not investment advice. Every result in this paper is a retrospective evaluation of what a fully-specified, mechanically- applied rule would have produced, under a real scoring methodology and a real external benchmark. It does not account for transaction costs, financing costs on short positions, capacity constraints at scale, or the operational requirements of actually deploying capital against these signals. Real-time validity in the sense argued throughout this paper – every decision uses only information available at the time – is a necessary condition for a result to be taken seriously, not a sufficient one for it to be tradeable.

9 Conclusion

A hierarchical spike-and-slab block model built to find genetic markers associated with a disease, with essentially no modification beyond re-thinking what "relevance" means for the new domain, produces a real, statistically significant edge in a genuinely adversarial financial forecasting competition – and that edge persists, undiminished, across three further years of out-of-sample data that had no influence whatsoever on any modeling choice in this paper. The methodological contribution we would emphasize most, though, is not the specific model: it is the discipline that made the result trustworthy in the first place. Every one of the three look-ahead leaks in Section 4 looked completely reasonable at the time it was introduced, and every one was only caught by the same mechanical question, applied retroactively – could this specific decision have been made with only the information available before the fact? A second application of the identical machinery, replicating an external benchmark rather than forecasting the same universe’s own returns, surfaced a genuinely more nuanced limitation than the one we expected to find: its risk is not that the tracked target stops performing, but that its return pattern stops being explainable by the chosen replicating universe – a subtler, and in this instance untriggered, failure mode. Both the substantive and the methodological results, we think, generalize well beyond this one competition.

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